

Cryptography sheet # 1

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In this sheet, we start playing with encryption techniques and the modular arithmetic.

Classroom exercises

1 Encryption and injectivity

A key property of an encryption function is that it must be reversible in a certain sense. Obviously, encryption that cannot be reversed is worthless. Interestingly, this concept of reversibility can be shown to correspond to the mathematical concept of injectivity. Verify this for yourself by proving the following statement:

Let X and Y be nonempty sets and e be a function from X to Y . The function e is injective if and only if there exists a function d from Y to X such that:

$$\text{for all } x \in X: \quad d(e(x)) = x.$$

2 Modular calculation for a double power

You are given a task to calculate $2026^{(2^{2026})}$ modulo 13. The first thing that comes to your mind is to calculate the double power explicitly and then do a long division. But then you do an internet search and realize that the estimated number of atoms in the universe is about 10^{80} .

That means, your double power has more digits than the number of atoms in the universe. That's a problem. You might not have enough paper to write it down. It will take quite a bit of time, too, and after all, you want to graduate soon. You think a bit more and realize that there are only 13 cosets of integers modulo 13. So, when you do modular calculations, the number of elements with which you calculate is quite limited. That is, when we do a power - the first, the second, the third and so on, things would get cyclic somehow... Can this be helpful to determine the answer efficiently?

Remark: we play with modular calculations to understand how they work and how to deploy them in cryptography. When we learn more, we'll have a mighty theory that allows us to solve such exercises in a routine manner. But when we try solving things without having theory, we prepare ourselves to perceiving the theory and we'll also be able to understand and appreciate the theory better.

Another remark: if you feel that a computer might help you to do some tedious modular calculations, go ahead and use it.

Additional exercises

3 Attack on Caesar

Imagine you are Asterix, and you intercept a message that Caesar sends via a courier to his legionnaires. The message is "exlbokuklvo". It does sound a bit like Latin, but actually does not make sense. You asked Getafix for advice and he suggested that this might be a cipher and the original message is in English. He also told you that it might be a good idea to use computer for breaking this code. Try to break the cipher and get the plain text.

4 Divisibility by 9

There is a rule that verifies divisibility of a non-negative integer number by 9 using the sum of its decimal digits. Discover this rule and explain why it works.

Remark: the key here is that 10 modulo 9 is very special.

Another remark: there is a completely analogous rule for divisibility by 3, too.